

SONG JIAN

Tokyo, Japan

✉ song@ms.k.u-tokyo.ac.jp jian.song@riken.jp

🏠 <https://jtrneo.github.io> [ResearchGate](#) [LinkedIn](#)

🎓 [Google Scholar](#) [GitHub](#) [ORCID](#)

Research Interests

- Remote sensing, 3D vision, high-fidelity synthetic 3D environments, advanced scene generation, transfer learning, scalable 3D understanding, and geospatial artificial intelligence.

Education

The University of Tokyo

Oct. 2021 – Jun. 2025

Ph.D. in Engineering

Chiba, Japan

- Laboratory: Machine Learning and Statistical Data Analysis Lab
- Supervisor: Prof. Naoto Yokoya
- Research topic: Synthetic data-driven 3D semantic reconstruction in remote sensing

Waseda University

Oct. 2019 – Mar. 2021

Master of Engineering

Fukuoka, Japan

- Major: Information Engineering
- Research topic: Flow-guided video object detection

Wuhan University

Sep. 2018 – Jun. 2021

Master of Engineering

Wuhan, China

- Major: Software Engineering
- Research topic: Remote sensing image semantic segmentation

Fujian Agriculture and Forestry University

Sep. 2011 – Jun. 2015

Bachelor of Agriculture

Fuzhou, China

- Major: Forestry Information Engineering

Employment History

RIKEN Center for Advanced Intelligence Project

Jul. 2025 – Present

Postdoctoral Researcher, Geoinformatics Team

Tokyo, Japan

- Advisor: Prof. Naoto Yokoya
- Research focus: high-fidelity synthetic 3D environments, advanced scene generation, transfer learning, remote sensing image synthesis, and scalable global 3D understanding.

RIKEN Center for Advanced Intelligence Project

Apr. 2021 – Jun. 2025

Research Part-timer, Geoinformatics Team

Tokyo, Japan

- Supervisor: Prof. Naoto Yokoya

The University of Tokyo

Oct. 2021 – May 2024

Research Assistant

Tokyo, Japan

- Supervisor: Prof. Naoto Yokoya

Inception Institute of Artificial Intelligence

Mar. 2019 – Aug. 2019

Research Intern

Abu Dhabi, UAE

- Supervisor: Dr. Fan Zhu

Land Satellite Remote Sensing Application Center

May 2018 – Sep. 2018

Research Intern

Beijing, China

- Supervisor: Dr. Yuhang Gan

Fundings and Awards

JSPS Grant-in-Aid for Early-Career Scientists (KAKENHI 26K21262) <i>PI; 4,550,000 JPY</i> – Project: “Open Synthetic Data Engine for Physics-Based Geospatial Foundation Models”	Apr. 2026 – Mar. 2031 [Link]
RIKEN BAIHO Award <i>Awarded for “Development of remote sensing image synthesis technology”</i>	Mar. 2025
AY2024 AI Center Fusion Research Promotion Fund <i>1,600,000 JPY</i>	Dec. 2023
SPRING GX Self-directed and Integrated Project Research <i>250,000 JPY</i>	Jul. 2024
SPRING GX Fellowship <i>180,000 JPY per month</i>	Apr. 2024
AY2023 AI Center Fusion Research Promotion Fund <i>2,000,000 JPY</i>	Dec. 2023
AY2023 GSFS Challenging New Area Doctoral Research Grant <i>600,000 JPY</i>	Mar. 2023
AY2020 JEES–SoftBank AI Scholarship <i>1,000,000 JPY</i>	Feb. 2020
Kitakyushu Science and Research Park Scholarship <i>300,000 JPY</i>	Sep. 2019

Publications (* denotes equal contribution)

5.1 Preprints

- H. Chen*, **J. Song***, W. Xuan*, J. Wang*, H. Qi, Z. Zhou, P. Dai, O. Dietrich, E. Gutierrez, L. Bromly, E. Nemni, Y. Ou, J. Zhao, Z. Zheng, Y. Xu, R. Hänsch, W. Jiao, M. Chini, C. Persello, J. Xia, S. Lu, L. Wang, Z. Zhu, E. Shelhamer, J. Chanussot, K. Schindler, and N. Yokoya, “Earth Observation for Disaster Mapping: Benchmarks, Methods, Challenges and Future Perspectives,” *Preprint*, 2026. [\[Paper\]](#) [\[Code\]](#)

5.2 Selected Journal Articles

- H. Chen, C. Lan, **J. Song**, D. Ibañez, J. Xia, K. Schindler, and N. Yokoya, “Multimodal Remote Sensing Change Detection: An Image Matching Perspective,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 233, pp. 487–501, 2026. [\[Link\]](#)
- **J. Song**, H. Chen, and N. Yokoya, “Enhancing Monocular Height Estimation via Sparse LiDAR-Guided Correction,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 232, pp. 155–171, 2026. [\[Link\]](#)
- H. Chen, **J. Song**, O. Dietrich, C. Broni-Bediako, W. Xuan, J. Wang, X. Shao, Y. Wei, J. Xia, C. Lan, K. Schindler, and N. Yokoya, “Bright: A Globally Distributed Multimodal Building Damage Assessment Dataset With Very-High-Resolution for All-Weather Disaster Response,” *Earth System Science Data*, vol. 17, no. 11, pp. 6217–6253, 2025. ([IEEE GRSS Data Fusion Contest 2025 official dataset](#)) [\[Link\]](#)
- J. Xia, H. Chen, C. Broni-Bediako, Y. Wei, **J. Song**, and N. Yokoya, “OpenEarthMap-SAR: A Benchmark Synthetic Aperture Radar Dataset for Global High-Resolution Land Cover Mapping,” *IEEE Geoscience and Remote Sensing Magazine*, vol. 13, no. 4, pp. 476–487, 2025. [\[Link\]](#)
- H. Chen*, **J. Song***, C. Han, J. Xia, and N. Yokoya, “ChangeMamba: Remote Sensing Change Detection With Spatiotemporal State Space Model,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–20, 2024. ([ESI Hot Paper & ESI Highly Cited Paper](#)) [\[Link\]](#)
- C. Broni-Bediako, J. Xia, **J. Song**, H. Chen, M. Siam, and N. Yokoya, “Generalized Few-Shot Semantic Segmentation in Remote Sensing: Challenge and Benchmark,” *IEEE Geoscience and Remote Sensing Letters*, pp. 1–5, 2024. [\[Link\]](#)
- H. Chen, C. Lan, **J. Song**, C. Broni-Bediako, J. Xia, and N. Yokoya, “ObjFormer: Learning Land-Cover Changes From Paired OSM Data and Optical High-Resolution Imagery via Object-Guided Transformer,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–22, 2024. [\[Link\]](#)

- X. Shao, H. Chen, K. Magson, J. Wang, **J. Song**, J. Chen, and J. Sasaki, “Deep Learning for Multi-Label Classification of Coral Conditions in the Indo-Pacific via Underwater Photogrammetry,” *Aquatic Conservation: Marine and Freshwater Ecosystems*, vol. 34, no. 9, e4241, 2024. ([Front cover article](#)) [[Link](#)]
- H. Chen, **J. Song**, C. Wu, B. Du, and N. Yokoya, “Exchange Means Change: An Unsupervised Single-Temporal Change Detection Framework Based on Intra- and Inter-Image Patch Exchange,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 206, pp. 87–105, 2023. [[Link](#)]

5.3 Selected Conference Papers

- J. Wang*, W. Xuan*, H. Qi, Z. Liu, K. Liu, Y. Wu, H. Chen, **J. Song**, J. Xia, Z. Zheng, and N. Yokoya, “DisasterM3: A Remote Sensing Vision-Language Dataset for Disaster Damage Assessment and Response,” *Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2025. [[Link](#)]
- **J. Song**, H. Chen, W. Xuan, J. Xia, and N. Yokoya, “SynRS3D: A Synthetic Dataset for Global 3D Semantic Understanding from Monocular Remote Sensing Imagery,” *Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2024. ([Spotlight Paper, top 3.08%](#)) [[Paper](#)] [[Project](#)] [[Code](#)]
- N. Yokoya, J. Xia, C. Broni-Bediako, **J. Song**, and H. Chen, “OpenEarthMap Benchmark Suite and Its Applications,” *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024. ([Oral Presentation](#)) [[Link](#)]
- H. Chen, **J. Song**, and N. Yokoya, “Change Detection Between Optical Remote Sensing Imagery and Map Data via Segment Anything Model (SAM),” *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024. ([Oral Presentation](#)) [[Link](#)]
- **J. Song**, H. Chen, and N. Yokoya, “SyntheWorld: A Large-Scale Synthetic Dataset for Land Cover Mapping and Building Change Detection,” *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024. [[Link](#)]
- **J. Song**, B. Adriano, and N. Yokoya, “Disaster Detection from SAR Images With Different Off-Nadir Angles Using Unsupervised Image Translation,” *2nd CDCEO Workshop at IJCAI*, 2022. [[Link](#)]

Academic Services

- Reviewer: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- Reviewer: IEEE Transactions on Image Processing (TIP)
- Reviewer: IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- Reviewer: Engineering Applications of Artificial Intelligence (EAAI)
- Reviewer: Conference on Neural Information Processing Systems (NeurIPS) 2025
- Reviewer: IEEE Transactions on Geoscience and Remote Sensing (TGRS)
- Reviewer: ISPRS Journal of Photogrammetry and Remote Sensing
- Reviewer: IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)
- Reviewer: IEEE Access
- Co-organizer: IEEE GRSS Data Fusion Contest 2025 [[Link](#)]
- Co-organizer: OpenEarthMap Land Cover Mapping Few-Shot Challenge at L3D-IVU CVPR 2024 Workshop [[Link](#)]